TerraTransfer

DATA LOGGER · TYP

Aquatos Web LTX.

GROUNDWATER · WATER LEVEL · ENVIRONMENTAL MONITORING

DESCRIPTION

The **Aquatos Web LTX** is TerraTransfer's flagship data logger for continuous, autonomous environmental monitoring. It captures field measurements and transmits them automatically via **LTE-M cellular** to the Sensormanager web portal — encrypted with **AES-128-CBC** (BSI-recommended) from the sensor interface all the way to the cloud.

Up to **24 digital channels (SDI-12)** allow direct connection of pressure sensors, multi-parameter probes, temperature chains, rain gauges and water meters. Local configuration is done through the browser via **Bluetooth 5** — no app installation required, secured with challenge-response authentication and a 6-digit PIN.

Designed for years of low-maintenance outdoor operation: **up to 10 years of battery life** (more than 3 million measurements / > 30,000 transmissions), **IP68**, operating range **−25 °C to +85 °C**, available as above-ground or in-well variant.

APPLICATIONS

- Groundwater and water-level monitoring — pipe networks, wells, resource tracking
- Surface waters — rivers, streams, reservoirs, tidal zones
- Flood early warning and level alarms
- Climate and weather stations (solar-autonomous)
- Multi-parameter water quality (pH, O_2, conductivity, turbidity and more)
- Regulatory monitoring networks and critical infrastructure
- Autonomous deployment in remote or harsh environments

TYPE 1500 · LTE-M

KEY FEATURES

- **LTE-M** with NB-IoT fallback, **Bluetooth 5** short-range
- **AES-128-CBC** end-to-end, BSI-compliant
- **Signed firmware updates** only (BLE or OTA)
- **24 × SDI-12** channels, bus wiring > 500 m
- Freely programmable sampling and transmission cycles
- **IP68**, ice-free operation −25 °C to +85 °C
- Above-ground and in-well housings, stainless steel
- Direct link to **Sensormanager web portal**

Technical Data & Accessories

AQUATOS WEB LTX · TYPE 1500

TECHNICAL SPECIFICATION

Measured parameters	Depending on connected sensor — pressure, water level, temperature, conductivity, pH, O_2, turbidity, nitrate, ammonium, phosphate, redox, chlorine, salinity, total hardness
Data transmission	LTE-M with NB-IoT fallback (bidirectional), Bluetooth 5 (BLE) local
Input channels	24 digital channels · SDI-12 V1.3 · up to 24 sensors on bus
Encryption	AES-128-CBC (BSI-recommended) · BLE auth: challenge-response with 6-digit PIN · fully encrypted firmware · signed updates only
Firmware	Type 1500 (LTX family) · OTA via LTE-M or local via BLE · signed updates only
Data handling	Automatic · freely configurable · real-time access via Sensormanager web portal · CSV / FTP export
Power supply	Lithium battery · up to 10 years operation · more than 3 million measurements / > 30,000 transmissions
Operating temperature	−25 °C to +85 °C · ice-free
Protection class	IP68
Enclosures	Above-ground (protective housing) · in-well (stilling-pipe mount)
Alarming	SMS / e-mail on threshold breach · self-diagnosis (battery, signal, temperature, humidity)
Compliance	CE (RED 2014/53/EU, EMC 2014/30/EU) · RoHS-compliant · BSI-secure web interface

Sensormanager

Web portal with database, time series, maps, alarms and API. Browser and mobile web app. Data retention up to 10 years, GDPR-compliant.

BLX.JS Dashboard

Browser-based configuration via Web Bluetooth — no app required. Chrome, Edge, Opera, Android. Open source (MIT). SDI-12 over BLE supported.

Compatible sensors

Piezo and ceramic pressure sensors, pulse / frequency counters (rainfall, water meters), Modbus converters, temperature chains, multi-parameter probes.

TerraTransfer

MADE IN GERMANY

ADDRESS

TerraTransfer GmbH
Ottostr. 19a
44867 Bochum · Germany

CONTACT

Phone +49 (0) 2327 83 44 85-1
info@terratransfer.de
terratransfer.de

REGISTRY

HRB 6778 · Bochum Court
VAT ID DE 209458277
CEO: M. Delker

SUBJECT TO CHANGE · WEIGHTS AND DIMENSIONS ARE REFERENCE VALUES

© 2026 TERRATRANSFER GMBH · VERSION 2026-04